FrameFlow

PRODUCT · DESIGN · ARCHITECTURE — V0.1

Long videos in. Viral desi shorts out.

The mobile-first AI clipping studio for Indian creators — Hinglish-native captions, clips in minutes, priced for Bharat.



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5 October 2026 · Confidential draft

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How to read this. Numbers marked *target* are goals to validate in Phase 0, not measurements. Competitor facts come from the teardown in `docs/research/competitive-teardown.md` (public sources, 5 Oct 2026). Mocks are high-fidelity concepts — final visual design will iterate with real content and user testing.

01 Product overview

1.1 The opportunity

India's creator economy runs on short video — Reels and YouTube Shorts — but the content that matters most is long-form: Hindi/Hinglish podcasts, talks, kathas, classes and debates. Turning one 60–120 minute episode into 20–50 shorts is today either manual editing (VN, InShot, CapCut-via-VPN) or Western tools that don't understand Indian speech.

No Hindi

OpusClip, the clipping leader, officially doesn't support Hindi or any Indic language; Android app has no editor (2.8★).

2–5 min

Kalakar, the desi-caption specialist, caps videos at 2–5 minutes and doesn't clip. No iOS app.

₹969+/mo

Captions (Mirage) has Indic clipping on mobile, but it's pricey, slow (up to ~1 hr) and its Android Lite app is 3.5★.

Platforms now give away *basic* captions and dubbing (Edits, Meta AI dubbing, YouTube Create). Value has moved to three places: **picking the right moments** for Indian audiences, **Indic caption craft**, and a **fast, cheap, reliable workflow** on the phones and networks Indian creators actually use.

1.2 Vision & north star

> **Turn a Hindi/Hinglish podcast into 20 ready-to-post clips — from a phone, in minutes, for less than a chai a day.**
>
> **North-star metric:** clips exported per week — captures selection quality, editor usability and willingness to pay in one number.

1.3 Who we serve (in order)

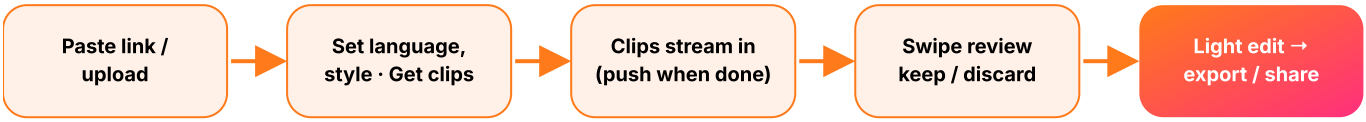
Persona	Job to be done	What wins them
The Clipper	Process many hours/week of podcasts into high-volume shorts across many accounts; paid per view (Whop and similar programmes).	Paste-a-link speed, batch, no watermark, cost per clip, multiple brands/accounts.
The Podcaster	Get 20–50 on-brand clips per episode without an editor (or make their editor 5× faster).	Great picks, speaker split layouts, accurate Hinglish captions, brand style.
The Editor	Deliver captioned shorts to clients fast with exact timing and style control.	Word-level caption control, presets, replace-source, SRT export.
Long-form creators	Repurpose long talks/classes/discourses into daily shorts.	Genre-aware picks (keep the punchline, the verse, the key concept), Devanagari captions.

1.4 Positioning

	OpusClip	Captions	Kalakar	VN	FrameFlow
AI clipping	✓ (20–40 min)	✓ (up to ~1 hr)	✗	✗	**✓ first clip <5 min (target)**
Hinglish / Indic	✗	✓	✓ (script locked)	Weak	**✓ instant script toggle**
Real mobile editor	iOS captions only; Android none	✓ (Lite on Android)	Web wrapper	✓ (manual)	**✓ Android-first**
Pricing model	Per source minute, credits expire	Credits, opaque	Hours + caps	Sub + credits	**Per exported clip, cost shown first, UPI**

02 Design mocks

2.1 Core flow



Target: link pasted → first reviewable clip in under 5 minutes for a 60-minute podcast; full set streams in behind it.

2.2 Design principles

1. Preview never waits for a render

Captions, cuts and reframing are drawn live on-device over the source. Rendering happens once, on export.

2. The video is always visible

Split layout: preview on top, tools below. Sheets never cover more than half the screen (Kalakar's biggest mobile flaw).

3. Hinglish is a first-class language

"Hinglish" is an explicit choice; script (Roman / देवनागरी / both) can be switched anytime without re-processing.

4. Money is never a surprise

Cost is shown before any processing. You pay for clips you export, not minutes you upload.

5. Explain the AI

Every clip shows *why* it was picked (hook, payoff, topic) — not just a score users don't trust.

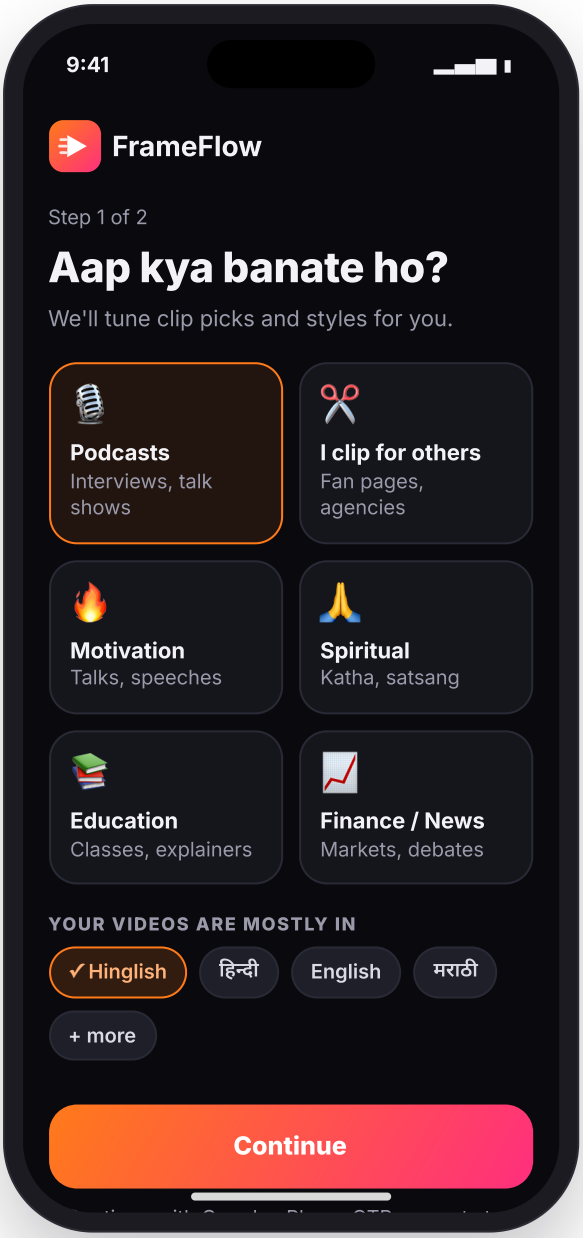
6. Built for real phones & networks

Works on 3–4 GB RAM Android, resumable everything, background processing, small app size.

Visual language: dark UI (creator-native, makes video pop), saffron→magenta "flow" gradient for primary actions, green/amber scores, Inter + Noto Sans Devanagari. Thumb-zone primary actions; one primary CTA per screen.

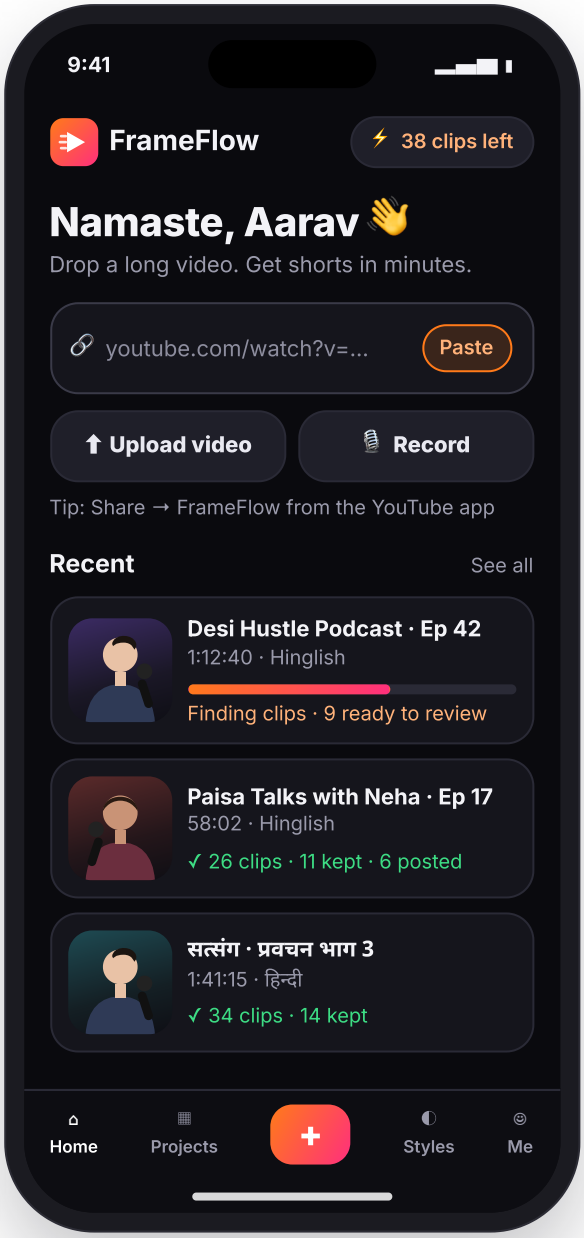
2.3 Screens

01 Onboarding — personalise in 2 taps



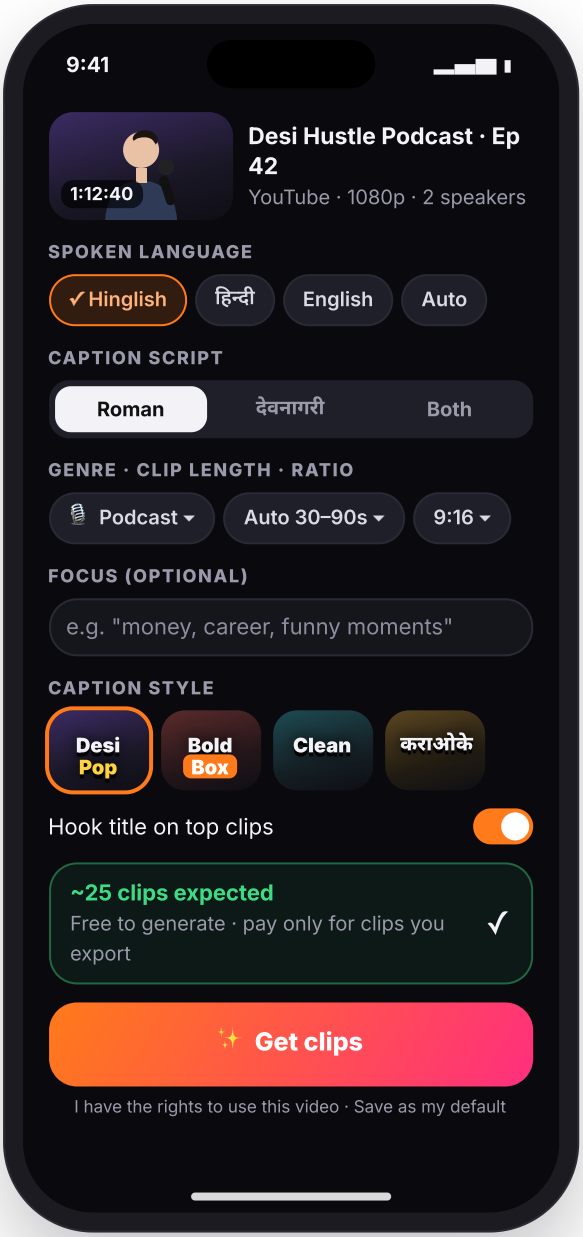
- **Persona + genre** pick sets default genre prompt, clip length and caption template.
- **Language first**: "Hinglish" is explicit (Kalakar makes users pick "English" for Hinglish — a known confusion).
- Sign-in after value is shown: Google or phone OTP (+91 default). No password.

02 Home — paste a link, done



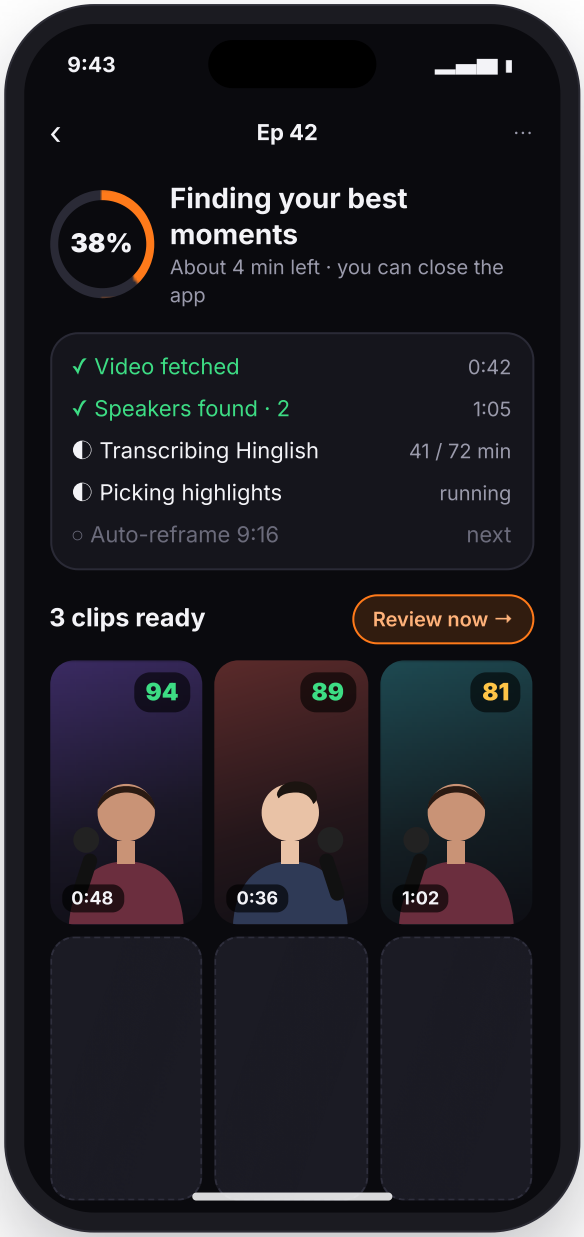
- **Link first**: no multi-GB upload over mobile data. Also a share-sheet target from YouTube.
- **Balance always visible** in clips, not credits/minutes.
- Projects show live status: clips ready while processing continues.

03 Get clips — one sheet, smart defaults



- Pattern from OpusClip's submit sheet + Kalakar's prepare modal, with **Hinglish** and **script** made explicit.
- **Cost shown before processing** — the category's #1 complaint is surprise credit burn.
- Rights confirmation line; defaults remembered per user.

04 Processing — clips stream in



- **Progressive results**: review the first clips while the rest process (OpusClip makes users wait 20–40 min).
- Honest step list + ETA; Android ongoing notification / iOS Live Activity mirror this.
- Failure ⇒ automatic, visible refund (nothing is charged until export anyway).

05 Results — ranked, explained



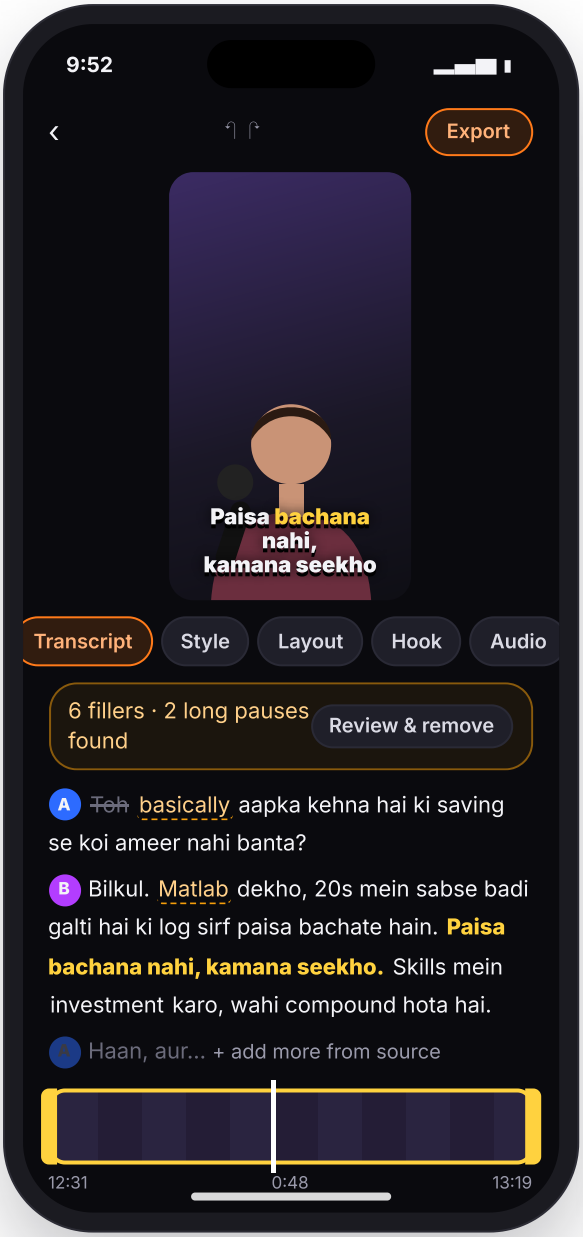
- **Score + plain-language reason + AI title** on every card (Vizard/OpusClip pattern; reason added because raw scores aren't trusted).
- Topic filter chips auto-extracted; **Reprompt** box asks for more clips without re-processing P1.
- Multi-select → bulk export.

06 Swipe review — Reels-style triage



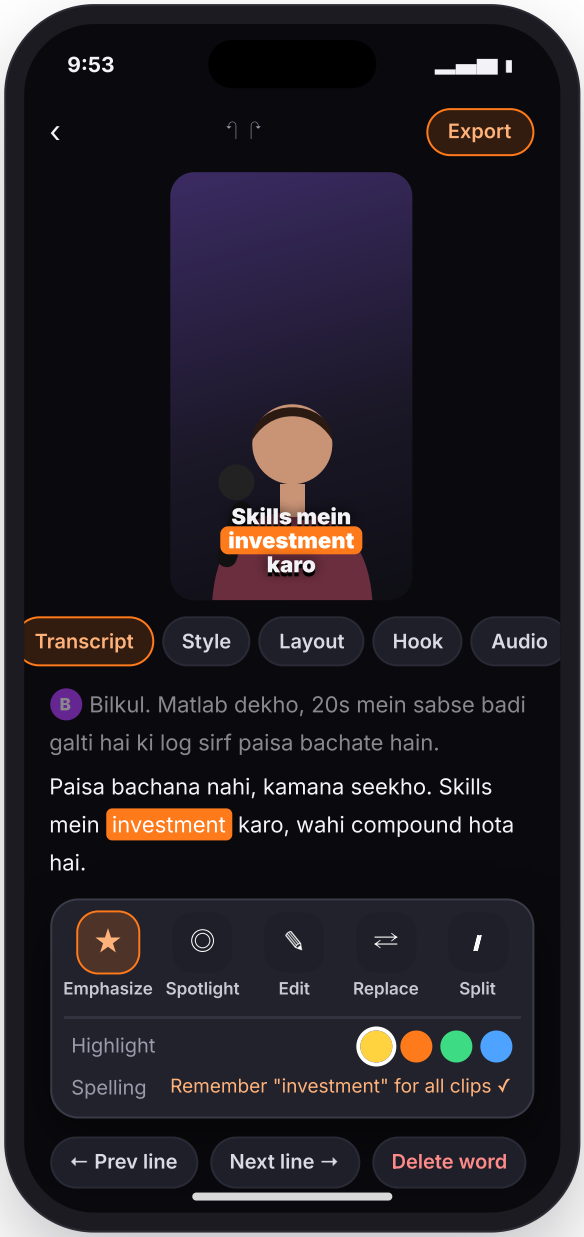
- **Our signature interaction**: full-screen vertical pager, the way users already consume Reels.
- Every keep/discard is training data for the India-tuned ranker (our moat).
- Script toggle right on the clip — no regeneration.

07 Editor — the transcript is the timeline



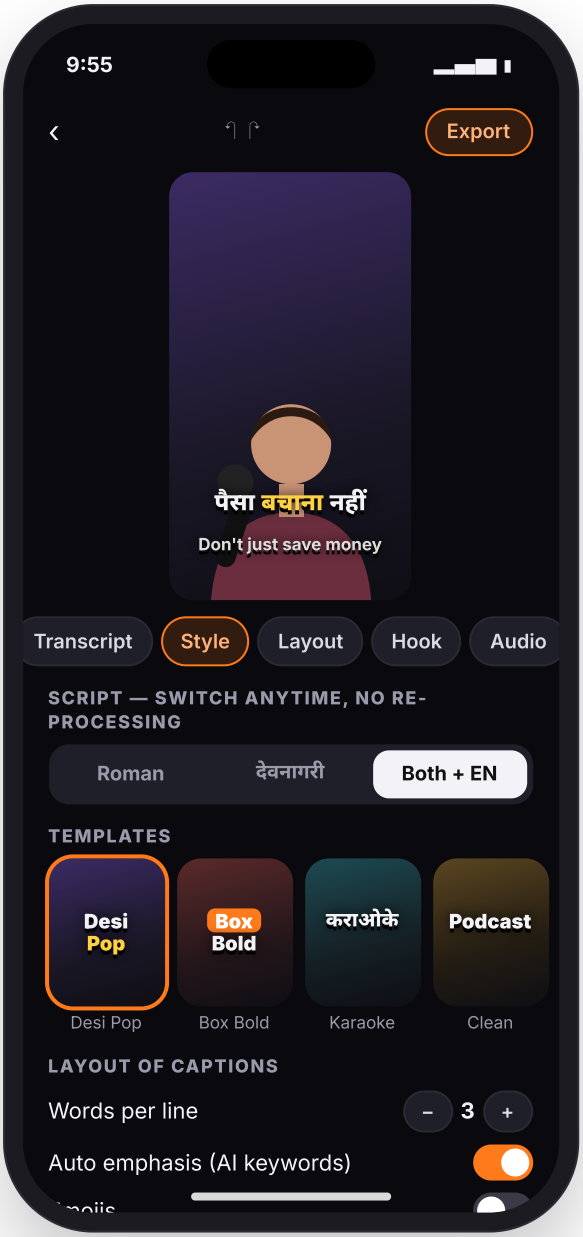
- **Delete words to cut** the video; struck-through words are removed. "+ add more from source" extends context.
- **Hindi filler detection** (matlab, basically, toh, na) with review-before-cut P1.
- Speaker chips from diarization; trim handles on mini-timeline for fine control.

08 Word tap menu — fix anything in one tap



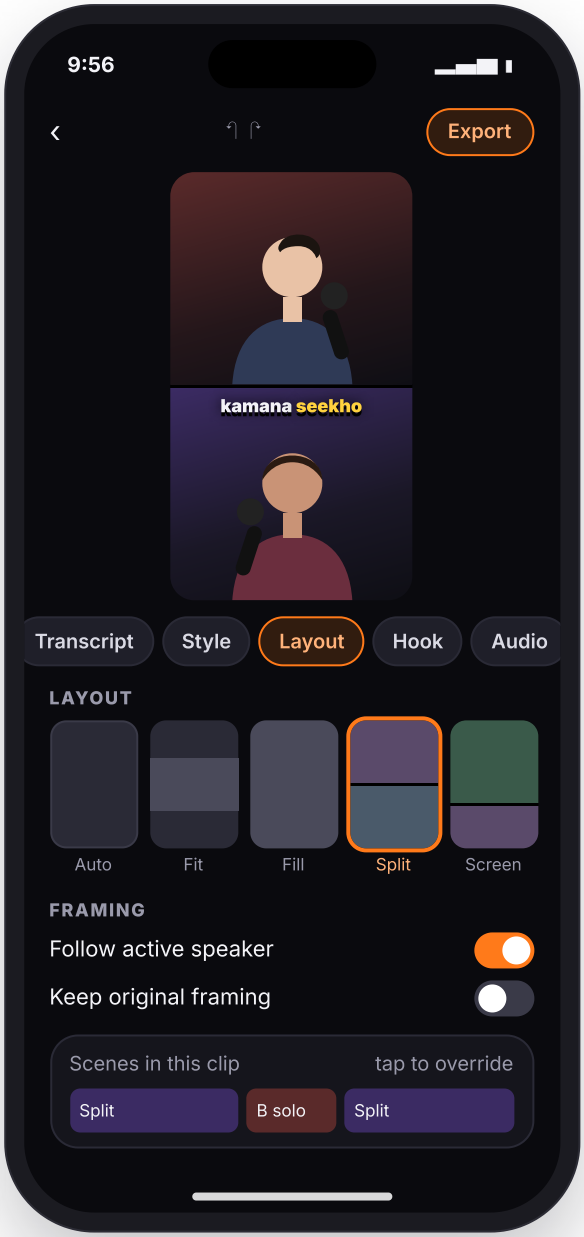
- Kalakar's best idea, made touch-native: **Emphasize · Spotlight · Edit · Replace all · Split**, move to prev/next line.
- **Remember spelling** → brand vocabulary (names, Hinglish spellings) applied to future transcripts.
- Preview stays visible; changes apply instantly (no render).

09 Style — script toggle + templates



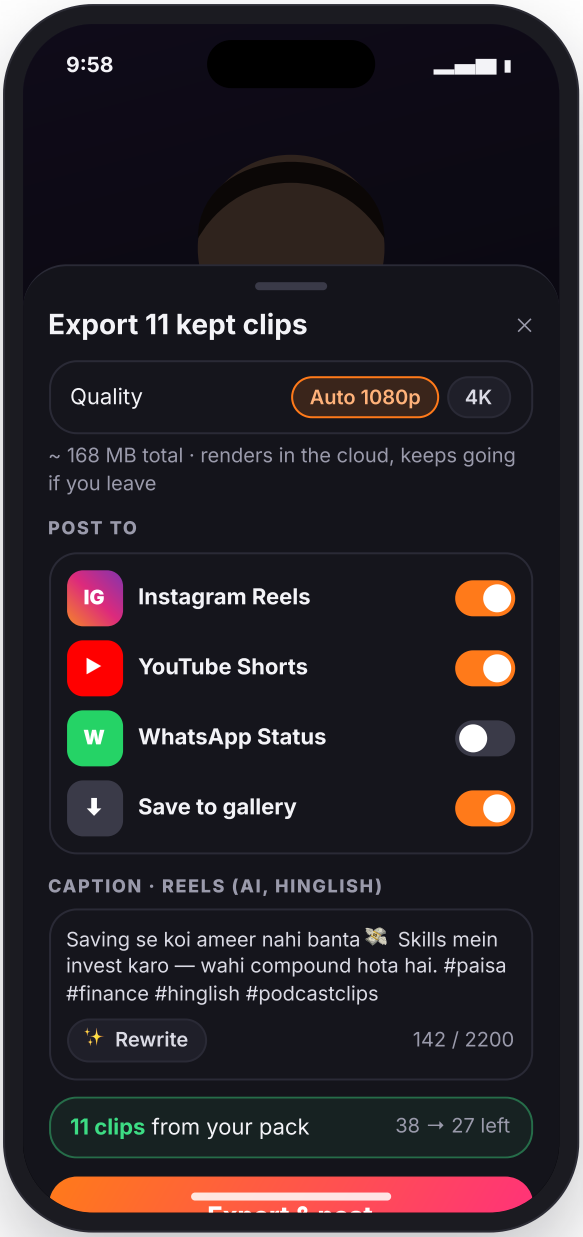
- **Instant script switch**: Roman, देवनागरी, or bilingual with English line — one transcript, three renderings.
- Indic-shaped fonts per script; **gesture positioning** on canvas (Kalakar uses numeric X/Y fields).
- "Apply to all kept clips" + save as brand preset P1.

10 Layout — speaker-aware reframe



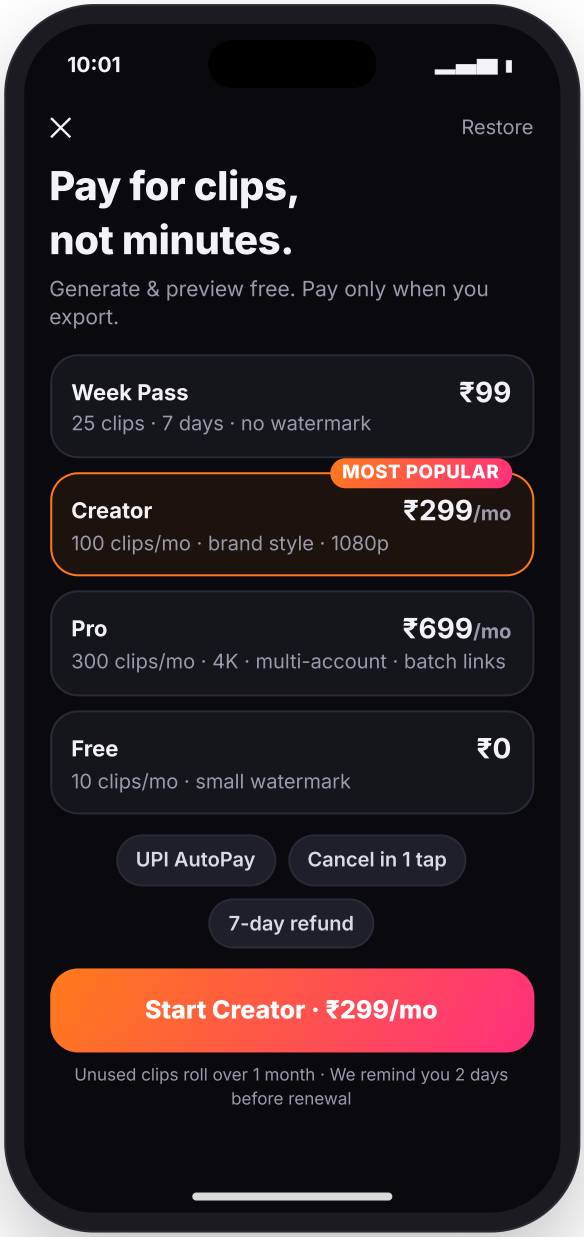
- **Split layout** for 2-person podcasts — the dominant Indian format; auto switches to solo when one person talks.
- **Keep original framing** escape hatch (reframe "fighting the shot" is a top OpusClip complaint).
- Per-scene overrides in a simple strip, not a full NLE timeline.

11 Export & share — per platform



- **Server render + background**: no "keep the app open" (a top VN/OpusClip complaint).
- Destinations that matter in India: **Reels, Shorts, WhatsApp Status**, gallery. AI caption & hashtags in Hinglish per platform.
- Charge = clips exported, shown before confirming.

12 Plans — pay for clips, not minutes



- Prices are **hypotheses** — Phase 0 measures unit cost; final packaging after beta.
- Trust features as the pitch: cost-first, roll-over, renewal reminder, refund window (answers the category's #1 complaint).
- Store billing rules apply (App Store IAP; Play Billing / user-choice billing in India) — see §3.6.

03 Product requirements (PRD)

Scope: FrameFlow v1 (MVP) on Android and iOS, plus the backend that powers it. Priorities: P0 required for MVP · P1 fast-follow (Phase 2) · P2 later.

3.1 Goals & non-goals

Goals (MVP)

- A creator can go from a YouTube link or upload of a Hindi/Hinglish/English video (up to 3 h) to exported, captioned 9:16 clips entirely on a phone.
- First reviewable clip in < 5 min for a 60-min source (target).
- Hinglish captions good enough that most clips need ≤ 2 word fixes.
- ≥ 50% of generated clips judged "postable as-is or with light edits" by beta users (target).
- Transparent pay-per-exported-clip pricing in INR with UPI.
- Profitable unit economics at ≤ ₹3 cost per exported clip (target, Phase 0 to verify).

Non-goals (v1)

- Generative video, AI avatars, text-to-video.
- Dubbing / voice cloning (platforms give it away).
- Deep multi-track NLE editing (VN owns this; we match gestures, not depth).
- Desktop/web editor (web is used only for marketing & account management at launch).
- Scheduling and direct API posting to every platform (share-sheet first; scheduling P1).
- Languages beyond Hindi, Hinglish, English in MVP (regional in Phase 3).

3.2 Success metrics

Metric	Definition	Beta target
North star — clips exported / week	Clips rendered for export or shared, deduplicated	Grows week over week
Activation	% of new users who export ≥ 1 clip within 24 h of sign-up	≥ 40%
Time to first clip (TTFC)	Submit → first clip visible in results (p50 / p90), 60-min source	< 5 min / < 8 min
Keep rate	Kept ÷ reviewed clips	≥ 35%
Caption edit rate	Word edits per exported clip	≤ 2
Pipeline success	Projects completing without error	≥ 98%
Retention	W1 / W4 retention of activated users	≥ 45% / ≥ 25%
Monetisation	Free → paid conversion of activated users (30 d)	≥ 6%
Unit cost	Infra + vendor cost ÷ exported clips	≤ ₹3
Quality	Crash-free sessions (Android / iOS)	≥ 99.5%

3.3 Key user stories

#	As a… I want… so that…	Pri
US1	As a **clipper**, I paste a YouTube link and get ranked clips without uploading anything, so I can work from my phone on mobile data.	P0
US2	As a **podcaster**, I choose Hinglish and Roman or Devanagari captions, and can switch script later without redoing the project.	P0
US3	As any user, I see the first clips within minutes and get notified when everything is ready, so I don't stare at a spinner.	P0

US4	As any user, I swipe through clips to keep or discard quickly, and understand why each was picked.	P0
US5	As an **editor**, I fix a wrong word in one tap and have the fix remembered for future videos.	P0
US6	As a podcaster, two speakers are shown in a split layout automatically, and I can override per scene.	P0
US7	As any user, I know what an export costs before I confirm, and pay with UPI.	P0
US8	As a creator, I trim by deleting words in the transcript and remove Hindi filler words after reviewing them.	P0 / P1
US9	As a clipper, I ask "3 more clips about investing" without reprocessing the video.	P1
US10	As an agency, I process a batch of links into several brands' presets and post to multiple accounts.	P1
US11	As a creator, I see which clips performed and get suggestions for what to clip next.	P2

3.4 Functional requirements

A. Onboarding & account

ID	Requirement	Pri
A1	Sign in with Google or phone OTP (+91 default); Apple sign-in on iOS. Guest preview allowed until first submit.	P0
A2	Persona + genre + primary language picker; sets defaults (genre prompt, clip length, template, script).	P0
A3	App UI in English and Hindi.	P1
A4	Account deletion + data export from settings (store & DPDP requirements).	P0

B. Ingest

B1	Paste a YouTube URL (public/unlisted) or use Share → FrameFlow from YouTube/Files/Gallery. Validate, show title, duration, thumbnail before submit.	P0
B2	Upload from gallery/files: chunked, resumable, continues in background; optional on-device compression for > 1080p sources.	P0
B3	Limits: up to 3 h and 10 GB per source in MVP; MP4/MOV/MKV/WebM; rights-confirmation line on submit.	P0
B4	Submit sheet: spoken language (Hinglish / हिन्दी / English / auto), caption script, genre, clip length, aspect ratio, optional focus prompt, caption template, toggles (hook title, clean audio, emojis), expected clip count, "save as default".	P0
B5	Auto-import new uploads from a connected YouTube channel.	P2

C. Processing

C1	Streaming pipeline: transcription and clip finding start before ingest completes; clips are pushed to the client as they are produced.	P0
C2	Live step list with ETA; Android ongoing notification and iOS Live Activity; push notification when complete or failed.	P0
C3	Word-level timestamps, speaker diarization, canonical transcript + Roman/Devanagari transliteration + English translation per word/line.	P0
C4	Clip selection: 30–90 s default (configurable 15 s–3 min), each with score (0–99), Hook/Flow/Value grades, plain-language reason, AI title, topic tags. Clips may stitch up to 3 segments.	P0
C5	Auto-reframe to 9:16 with active-speaker tracking; layouts Fill, Fit, Split (2-up), Screenshare; per-scene layout decisions.	P0
C6	Failure handling: automatic retry per stage; clear error reasons; nothing charged (charges only on export).	P0
C7	Optional audio clean-up (denoise) before transcription.	P1

D. Review

D1	Results grid sorted by score (toggle: timeline order); topic filter chips; multi-select.	P0
D2	Full-screen swipe review: keep / discard / edit; shows score, grades, reason, source timestamps, speaker; script toggle.	P0
D3	Reprompt: natural-language request for more/different clips using the cached transcript (no re-ingest, no re-ASR).	P1

E. Clip editor

ID	Requirement	Pri
E1	Split layout: live preview (top) + tool tabs (Transcript, Style, Layout, Hook, Audio). Preview never hidden by more than 50%. Undo/redo always visible.	P0
E2	Transcript editing: delete words/sentences to cut video; restore deleted; "add more from source" to extend start/end; trim handles on mini-timeline (0.1 s precision).	P0
E3	Word tap menu: Emphasize, Spotlight, Edit text, Replace all, Split line, move to previous/next line, delete, highlight colour, "remember spelling" (brand vocabulary).	P0
E4	Caption styles: ≥ 10 templates in MVP with Indic-shaped fonts; word-by-word highlight (karaoke); words per line; position & scale by drag/pinch on canvas; auto-emphasis toggle; emoji toggle.	P0
E5	Script switch at any time: Roman / Devanagari / bilingual (+ English line). No re-processing.	P0
E6	Layout tab: Auto, Fit, Fill, Split, Screenshare; "follow active speaker"; "keep original framing"; per-scene override strip.	P0
E7	Hook title overlay (AI-suggested Hinglish options, editable, style presets).	P0
E8	Filler-word & long-pause removal for Hindi/Hinglish/English with review-before-cut list.	P1
E9	Brand presets (fonts, colours, template, logo, layout) applied at submit or to all kept clips.	P1
E10	Music bed, auto zoom punch-ins, B-roll.	P2

F. Export & share

F1	Server-side render of kept clips (1080p default, 720p option; 4K on Pro if source ≥ 4K) continuing in background; download to gallery.	P0
F2	Share to Instagram Reels, YouTube Shorts, WhatsApp Status via native share intents; AI caption + hashtags per platform (Hinglish), copy button.	P0
F3	Bulk export of selected clips; SRT/VTT export of captions.	P0 / P1
F4	Direct posting via platform APIs, multi-account, scheduling calendar.	P1

G. Billing

G1	Clip-based balance: generation & preview free; one clip consumed per exported clip (re-exports of the same clip free for 30 days).	P0
G2	Plans: Free, Week Pass, Creator, Pro (see §3.6). Cost shown before export; renewal reminder 2 days ahead; one-tap cancel; 7-day refund on unused packs.	P0
G3	Payments compliant per platform: App Store IAP on iOS; Google Play Billing (and user-choice billing with UPI where permitted) on Android; web checkout via Razorpay (UPI AutoPay) for account management.	P0
G4	Transparent usage ledger in Settings (every charge and refund).	P0

H. Support & trust

H1	In-app help + human support on WhatsApp (Hindi/English); feedback on any clip (👍/👎 with reason).	P0
H2	Privacy: sources and renders auto-deleted after retention window (default 30 days, configurable); never used to train models without opt-in.	P0

3.5 Non-functional requirements

Area	Requirement (targets)
Speed	TTFC p50 < 5 min, p90 < 8 min for a 60-min source; full project < 15 min p90. Editor interactions < 100 ms; script switch < 300 ms. Export of a 60 s clip < 60 s p90.
Devices	Android 10+ with 3 GB RAM; iOS 16+. Smooth 30 fps preview on a ₹12k Android phone. Install size < 60 MB.
Networks	Works on unstable 4G: resumable uploads/downloads, idempotent API calls, offline-tolerant editing (edits sync when back online).
Reliability	Pipeline success ≥ 98%; crash-free sessions ≥ 99.5%; API availability 99.9%; no data loss on app kill.
Caption quality	Hinglish WER and timestamp accuracy measured on the FrameFlow eval set; launch bar set in Phase 0 after the vendor bake-off.
Indic rendering	Correct shaping of conjuncts and matras (HarfBuzz) identical in preview and export; no tofu glyphs.
Security & privacy	TLS everywhere; encryption at rest; signed, expiring media URLs; least-privilege service access; compliance with India's DPDP Act; store privacy labels accurate.
Accessibility	Dynamic type, contrast ≥ 4.5:1 for text, screen-reader labels on all controls, haptics on key actions.
Observability	Per-stage pipeline timings and cost per project; client crash/ANR reporting; product analytics (see 3.7).

3.6 Pricing & packaging (hypothesis)

Plan	Price	Includes	Purpose
Free	₹0	10 exported clips / month, small watermark, 1080p	Activation, word of mouth
Week Pass	₹99 / 7 days	25 clips, no watermark	Low-commitment first purchase
Creator	₹299 / month	100 clips, brand preset, roll-over 1 month	Podcasters & creators
Pro	₹699 / month	300 clips, 4K, batch links, multi-account, priority queue	Clippers & editors
Agency	Custom	Team seats, many brands, API	Phase 2+

Store considerations: digital goods inside the iOS app use Apple IAP; on Android, Google Play Billing with user-choice billing (where available in India) to offer UPI. Store fees (15–30%) are included in the unit-economics model. Prices finalised after Phase 0 cost measurement and beta willingness-to-pay tests.

3.7 Analytics (core events)

`onboarding_completed` · `source_submitted` (type, duration, language, genre) · `first_clip_ready` (latency) · `project_completed` / `project_failed` (stage, reason) · `clip_reviewed` (kept/discarded, position, score) · `caption_word_edited` · `script_switched` · `layout_changed` · `export_started` / `export_completed` · `share_target_selected` · `paywall_viewed` · `purchase_completed` · `refund_requested`. Keep/discard and post-performance signals feed the ranking model (with consent).

3.8 Release plan

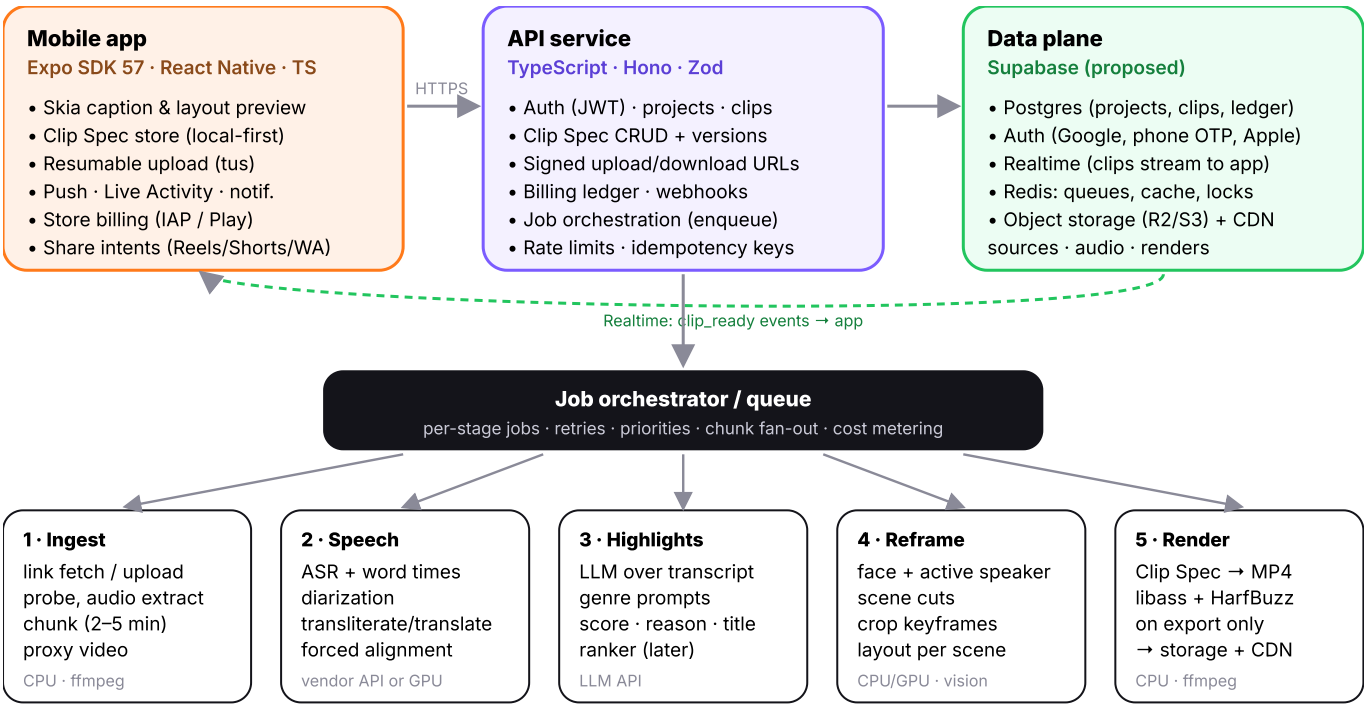
Phase	Duration	Scope	Exit criteria
0 · Prove it	~2 weeks	Hinglish/Hindi eval set (~50 h); ASR vendor bake-off; end-to-end pipeline spike (link → clips); caption render parity prototype; cost model.	ASR chosen; TTFC < 5 min on 60-min source; cost/clip ≤ ₹3 path; render parity proven.
1 · MVP	~6–8 weeks	All P0 requirements; Android + iOS internal builds via EAS; closed beta with 30–50 clippers & podcasters.	Activation ≥ 40%, keep rate ≥ 35%, pipeline success ≥ 98%, crash-free ≥ 99.5%.
Public launch	After beta	Store listings (Play first, App Store same week), pricing live, Hindi tutorials, creator affiliate programme.	Store approval; support load manageable.
2 · Retention	Ongoing	Reprompt, filler removal, brand presets, direct posting & scheduling, agency/batch mode, Hindi UI.	W4 retention ≥ 25%; paid conversion ≥ 6%.
3 · Expand	Later	Regional languages (Marathi, Bengali, Tamil, Telugu, Gujarati, Punjabi…), insights loop, learned ranker.	Ranker beats LLM-only keep rate.

3.9 Risks & open questions

Risk / question	Mitigation / decision needed	Owner
YouTube link ingest — ToS, cloud IP blocking, clippers using others' content	Rights confirmation; upload fallback; respect takedowns; decide policy on clipping third-party content.	Founder (policy) · Tech
Unit economics	Phase 0 measures cost per stage; render only on export; cache transcripts; pick cost-efficient ASR.	Tech
Hinglish ASR quality	Data-driven vendor choice on our eval set; brand vocabulary; fast word-fix UX.	Tech
Platform risk (YouTube/Meta free clipping)	Cross-platform workflow, agency features, India-tuned selection quality.	Founder · Tech
Store billing	IAP + Play Billing; validate user-choice billing for UPI; model store fees.	Tech · Founder
Open: cloud provider & budget	Recommendation in §4.9; needs sign-off.	Founder
Open: bundle ID, brand assets, legal entity for payments	Needed before store submission.	Founder

04 Technical architecture

4.1 System overview



External services

ASR vendor(s)
chosen by Phase 0 bake-off
Sarvam · ElevenLabs · Deepgram
Google · self-hosted Whisper

LLM
Claude (selection, titles, hooks, captions, reprompt)

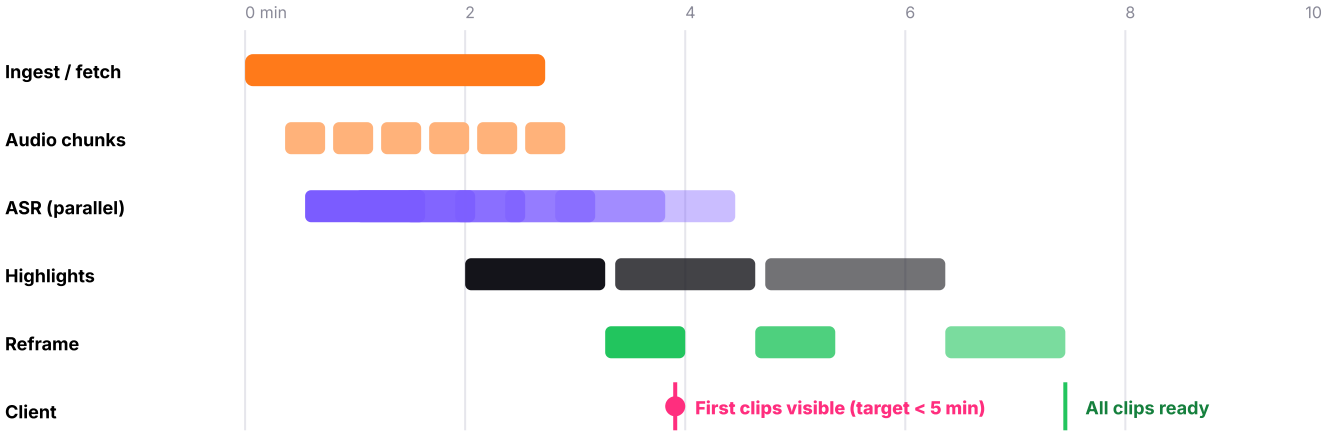
Payments
App Store IAP · Play Billing
Razorpay (UPI AutoPay, web)
RevenueCat (entitlements)

Platform services
Expo push · FCM · APNs
EAS Build/Update · Sentry
PostHog analytics

Vendor names in "External services" are candidates. Final choices are logged in §4.11 once measured. The system is designed to be vendor-swappable at each stage via a provider interface.

4.2 The clip pipeline (streaming)

A 60-minute source is split into overlapping audio chunks as soon as bytes arrive. Each chunk is transcribed independently; the highlight stage runs over a sliding window of finished transcript (with context from neighbours), so the first clips appear long before the whole episode is processed.



Illustrative timing for a 60-min Hinglish podcast; real numbers come from the Phase 0 spike.

Stage details

Stage	What it does	Key choices
Ingest	Fetch link or assemble resumable upload; ffprobe; extract 16 kHz mono audio; cut into 2–5 min chunks with overlap; create 360p/540p proxy for client preview.	Proxy streamed via CDN (HLS) so the phone never downloads the original.
Speech	Per-chunk ASR with word timestamps; merge overlaps; diarize speakers; forced alignment for tight caption timing; produce canonical script + Roman + Devanagari + English layers.	Provider interface; transliteration via model/LLM; brand vocabulary as ASR hints and post-correction.
Highlights	Sliding window over transcript (+ speaker turns, pauses, laughter) → candidate segments with start/end on word boundaries, score, Hook/Flow/Value, reason, title, tags; de-duplicate; enforce coverage across the episode.	LLM with genre prompts + structured output; later a learned ranker on keep/discard + post performance.
Reframe	Sample frames; face detection + tracking; active-speaker via audio–visual correlation; scene cut detection; emit crop keyframes and per-scene layout (solo/split/fit).	Runs only on candidate clip ranges, not the full video (cost).
Render	On export: cut source ranges, apply crop/layout, burn captions (ASS via libass + HarfBuzz) and overlays, encode H.264/AAC, upload, notify.	Deterministic from Clip Spec → identical to preview; cached by spec hash.

4.3 Clip Spec — the single source of truth

Every clip is a versioned JSON document. The app renders it live (Skia over the proxy video); the render worker turns the same document into an MP4. Editing, undo/redo, templates, presets and reprompts are all operations on this document.

```
// ClipSpec v1 (abridged)
{
  "id": "clp_7Hk2", "projectId": "prj_42", "version": 14,
  "aspect": "9:16",
  "segments": [ { "src": "src_main", "in": 751.20, "out": 799.48 } ],      // up to 3 stitched ranges
  "cuts": [ { "from": 752.10, "to": 752.62, "reason": "filler" } ],      // word deletions
  "captions": {
    "script": "roman",            // roman | devanagari | bilingual
    "templateId": "desi-pop", "wordsPerLine": 3, "position": { "x": 0.5, "y": 0.72, "scale": 1.0 },
    "words": [ { "t0": 753.04, "t1": 753.31, "spk": "B",
                 "text": { "roman": "paisa", "devanagari": "पैसा", "en": "money" },
                 "style": { "emphasis": true, "color": "#FFD23F" } } ],
    "lineBreaks": [ 3, 7, 11 ], "overrides": { "line:2": { "templateId": "box-bold" } }
  },
  "layout": { "mode": "auto", "followSpeaker": true,
    "scenes": [ { "t0": 751.2, "t1": 770.0, "type": "split", "crops": ["cropA", "cropB"] } ],
    "cropTracks": { "cropA": [ { "t": 751.2, "x": 0.31, "y": 0.40, "w": 0.32 } ] } },
  "hook": { "text": "Paisa bachana nahi, kamana seekho", "style": "banner-top", "until": 3.0 },
  "audio": { "denoise": false, "music": null },
  "ai": { "score": 92, "grades": { "hook": "A", "flow": "A-", "value": "B+" },
          "reason": "Contrarian money take, payoff in 41s", "tags": ["money", "career"] }
}
```

Why a spec, not a timeline file

- Instant, free edits (no server round-trip to see a change).
- Preview/export parity: one schema, two renderers, golden-image tests.
- Render caching by spec hash; re-exports cost nothing.
- Script switch = change one field; all three layers already stored.

Sync model

- Local-first: edits applied to a local copy (MMKV/SQLite), queued, synced with optimistic concurrency (`version`).
- Server stores full versions for undo history & support.
- Conflicts (rare, single user) resolved last-writer-wins per field group.

4.4 Mobile app architecture

Concern	Choice	Notes
Framework	Expo SDK 57, React Native 0.86, TypeScript, Expo Router (file-based), React Compiler	Already scaffolded in repo; dev builds (not Expo Go) because of native modules.
Video playback	`expo-video` playing CDN HLS proxy; frame-accurate seeking	Never download the original to the phone.
Caption/overlay preview	`@shopify/react-native-skia` canvas synced to player clock; Indic text via Skia Paragraph (HarfBuzz shaping)	Parity tests against server render.
Gestures & animation	react-native-gesture-handler, Reanimated 4 (worklets)	Drag/pinch captions, swipe review pager, trim handles.
State	Zustand (editor/Clip Spec with undo stack) + TanStack Query (server state)	Command pattern for undo/redo.
Local storage	MMKV for prefs/drafts; SQLite (expo-sqlite) for offline edit queue	Edits survive app kill.
Uploads	tus resumable uploads with background tasks; on-device compression (expo-video / native transcoder) for > 1080p	Resume across network drops and restarts.
Realtime & notifications	Supabase Realtime subscription per project; expo-notifications (FCM/APNs); Android foreground-service notification; iOS Live Activity (native module)	Progressive clip delivery.
Share	Native share intents to Instagram/YouTube/WhatsApp; save to gallery (expo-media-library)	Direct posting APIs in Phase 2.
Payments	RevenueCat (IAP + Play Billing entitlements); server-side ledger is the source of truth	User-choice billing / UPI to validate.
Quality	Sentry (crash/ANR), PostHog (analytics, flags), EAS Build/Submit/Update, Maestro E2E, Jest unit tests	OTA updates for JS fixes.
i18n	English + Hindi UI strings (i18next); Noto Sans + Noto Sans Devanagari bundled	Regional scripts later.

App structure (planned)

```
src/
  app/                # Expo Router routes
    (onboarding)/     # persona, language, sign-in
    (tabs)/           # home, projects, styles, me
    project/[id]/     # processing, results, review (swipe)
    clip/[id]/        # editor: transcript, style, layout, hook, audio
    export/           # export & share sheet
    paywall.tsx
  features/           # ingest, review, editor, export, billing (UI + hooks)
  engine/             # clip-spec schema (zod), ops, undo stack, skia renderer
  services/           # api client, realtime, uploads, notifications, analytics
  ui/                 # design system: tokens, buttons, sheets, chips
```

4.5 Backend services

Service	Responsibility	Tech
API	Auth verification, projects/sources/clips/specs CRUD, signed URLs, billing ledger & entitlements, webhooks (payments, providers), job enqueueing, rate limiting.	TypeScript, Hono, Zod, Drizzle ORM, Postgres
Orchestrator	Durable per-project workflow: stage dependencies, chunk fan-out/fan-in, retries with backoff, timeouts, cost metering, priority queues (Pro).	BullMQ on Redis (MVP) → Temporal if workflows outgrow it
Ingest worker	Link fetch, upload assembly, ffprobe, audio extraction, chunking, proxy HLS.	Python/Node + ffmpeg, CPU
Speech worker	Calls ASR provider(s) per chunk, merges, diarizes, aligns, transliterates/translates.	Python; provider adapters
Highlight worker	Windowed LLM selection with structured outputs; dedupe; scoring; titles, hooks, tags, social captions; reprompt handling.	Python/TS; Claude API
Vision worker	Face detection/tracking, active speaker, scene cuts, crop keyframes for candidate ranges.	Python; ONNX/MediaPipe; CPU first, GPU if needed
Render worker	Clip Spec → ASS subtitles + ffmpeg filter graph → MP4; upload; cache by spec hash.	ffmpeg + libass + HarfBuzz, CPU autoscaling
Notifier	Realtime events, push notifications, emails (receipts).	Supabase Realtime, Expo Push

4.6 Data model (core tables)

Table	Key fields
users	id, phone/email, persona, genres[], primary_language, default_settings jsonb, created_at
workspaces / members	id, owner_id, name, plan, brand_presets jsonb · (workspace_id, user_id, role) — agency mode
projects	id, workspace_id, source_type (link/upload), source_url, title, duration_s, language, script, genre, settings jsonb, status, stage_progress jsonb, error, created_at
media_assets	id, project_id, kind (original/audio/proxy/chunk/render), storage_key, bytes, codec, expires_at
transcripts	project_id, version, words (jsonb or columnar file in storage), speakers, layers (roman/deva/en), vendor, cost
clips	id, project_id, rank, score, grades, reason, title, tags[], status (new/kept/discarded/exported), spec jsonb, spec_version, prompt_id
clip_spec_versions	clip_id, version, spec jsonb, created_by, created_at
exports	id, clip_id, spec_hash, resolution, status, storage_key, cost_clips, completed_at
ledger	id, workspace_id, delta_clips, reason (grant/purchase/export/refund/rollover), ref_id, created_at — append-only
subscriptions / purchases	workspace_id, store (apple/google/razorpay), product_id, status, period_end, raw jsonb
vocabulary	workspace_id, term, preferred_roman, preferred_deva — brand spellings
feedback / events	clip_id, action (keep/discard/edit/post), metadata — training signal (consented)

4.7 API (v1, abridged)

Endpoint	Purpose
POST /v1/sources/resolve	Validate link → title, duration, thumbnail, est. clips
POST /v1/uploads	Create resumable (tus) upload; returns upload URL
POST /v1/projects	Create project from source + settings (Idempotency-Key)
GET /v1/projects/:id	Status, stage progress, ETA
GET /v1/projects/:id/clips	Ranked clips (cursor paging, filters)
POST /v1/projects/:id/reprompt	"More clips about…" using cached transcript
PATCH /v1/clips/:id	Status keep/discard; spec patch (JSON Patch + version)
GET /v1/clips/:id/spec?v=	Fetch spec version
POST /v1/exports	Export clip ids → charge ledger, enqueue render
GET /v1/exports/:id	Render status + signed download URL
GET /v1/me/balance · /v1/me/ledger	Clip balance, full usage history
POST /v1/webhooks/{revenuecat,razorpay}	Entitlement & payment events
realtime: project:{id}	stage_progress, clip_ready, project_done, export_done

4.8 ML strategy & evaluation

ASR bake-off (Phase 0)

- **Eval set:** ~50 h of real Hinglish/Hindi podcasts, talks, katha, news; 2–3 speakers; varied mics and noise. Human-verified transcripts for a 5 h subset.
- **Metrics:** WER/CER (Roman & Devanagari), code-switch accuracy, proper-noun accuracy, word-timestamp error (ms), diarization error rate, latency, cost/hour.
- **Candidates:** Sarvam, ElevenLabs Scribe, Deepgram, Google Chirp, self-hosted Whisper large-v3 variants / AI4Bharat models.

Clip selection

- **v1:** LLM over windowed, diarized transcript with genre prompts; structured output; rules for boundaries (complete thought, no cut punchline, hook in first 3 s).
- **Offline eval:** overlap with human-picked clips on the eval set; "postable" rate from blind reviewers.
- **Online:** keep rate, export rate, post performance.
- **v2:** learned ranker on consented keep/discard + views data → India-tuned virality model (moat).

Transliteration & translation

- Every word stored in Roman, Devanagari and (line-level) English.
- Consistent Hinglish spellings via normalisation table + brand vocabulary.

Reframe

- Face detection + tracking with temporal smoothing; active speaker by lip-motion/audio correlation.
- Eval: % frames with speaker in safe area; jitter metric; manual review set incl. handheld & gaming footage.

4.9 Infrastructure, cost & scale

Proposed stack (needs sign-off)

Layer	Proposal	Why
Database, auth, realtime	Supabase (Postgres + Auth + Realtime), Mumbai region	Fast to build; Realtime fits progressive clips; team already uses Supabase.
Object storage + CDN	Cloudflare R2 + CDN (or S3 + CloudFront)	Video egress is the big cost; R2 has no egress fees.
API	Containers on Fly.io / Cloud Run / ECS in Mumbai	Low latency to Indian users.
Workers	Autoscaling container pools (CPU for ingest/render; GPU pool only if self-hosting ASR/vision)	Scale to zero off-peak; spot instances for render.
Queue	Redis (Upstash/managed) + BullMQ	Simple, proven; migrate to Temporal if needed.
Mobile delivery	EAS Build / Submit / Update	Matches Expo stack; OTA fixes.

Unit cost model (to be filled by Phase 0)

Cost per exported clip = (ingest + ASR + LLM selection + reframe on candidates) ÷ clips exported per project + render + storage/egress + store fee share. Levers: render only on export, reframe only candidate ranges, cache transcripts (reprompt is LLM-only), proxy streaming instead of originals, retention auto-delete. **Target ≤ ₹3 per exported clip** so that ₹299 for 100 clips carries healthy gross margin after store fees.

Scale assumptions (beta → launch)

Assumption	Beta	Launch +3 months (illustrative)
Hours of source processed / day	50	2,000
Concurrent projects (peak)	10	300
Clips exported / day	500	20,000
Storage (30-day retention)	~2 TB	~80 TB

4.10 Security & privacy

- TLS 1.2+ everywhere; at-rest encryption for DB and storage; secrets in a managed vault; no secrets in the app bundle.
- Short-lived signed URLs for all media; per-workspace access checks on every object; Postgres row-level security.
- Data retention: originals and renders auto-deleted after 30 days (configurable); transcripts kept for re-editing until project deletion; account deletion purges within 30 days.
- User content is not used for model training without explicit opt-in; vendor DPAs reviewed (ASR, LLM).
- DPDP Act readiness: consent records, purpose limitation, grievance contact; accurate App Store/Play data-safety labels.
- Abuse & rights: rights confirmation, takedown process, rate limits per account/IP, payment fraud checks.

4.11 Decision log

Date	Decision	Status
2026-10-05	Mobile stack: Expo SDK 57, React Native 0.86, TypeScript, Expo Router.	Done
2026-10-05	Strategy: Hinglish podcasts + clippers wedge; Android-first with iOS in parallel; pay per exported clip.	Adopted
2026-10-05	Architecture: Clip Spec as source of truth; on-device preview, render only on export; streaming pipeline.	Proposed
Phase 0	ASR vendor, LLM prompts/models, reframe approach, caption renderer parity.	Pending data
Founder	Cloud provider & budget; YouTube-link policy for clippers; bundle ID; payments entity.	Pending

Appendix — competitive evidence (summary)

Competitor	Key facts used in this document
OpusClip	Help center: Hindi not supported; 25 non-Indic languages; iOS editor captions-only; Android no editor, 2.8★; ClipAnything 20–40 min; free projects expire in 3 days; credits per source minute.
Captions (Mirage)	Hindi, Hinglish + 10 Indic languages; AI Shorts up to ~1 hr; India pricing ₹599/wk, ₹969–2,999/mo; Android Lite 3.5★ with 5-min limit; opaque credit complaints.
Kalakar	21 languages with Native/Roman; captioning only; 2–5 min video caps (Free/Editor/Creator); Android app since Jul 2026 (~5K installs); no iOS; best-in-class word menu; mobile sheets hide preview.
VN	#2 Photo & Video on India App Store; 2.1M India ratings; no content-aware clipping; English-first captions; top complaints: audio on export, export failures, crashes; foreground-only export.
Market	CapCut banned in India; Josh stopped uploads; Moj pivoted to dramas; Meta/YouTube free captions & dubbing; Hindi "podcast shorts" tutorials at 4–5M views; Whop clipping tutorials 376K–676K views.

Full sources: `docs/research/competitive-teardown.md`. Strategy: `docs/strategy.md`.